The Reconstructed Theory Has One Mass: a Machine-Checked Volume-Uniform Spectral Gap with Exact Identification Against the Gibbs Sums

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Abstract

For the spatial $\mathbb{Z}_2$ (Ising-slice) system in the Dobrushin window $2\tanh|\beta| + 2\tanh|\gamma| \leq \alpha < 1$, we machine-check in Lean 4 an end-to-end chain: (i) the Osterwalder–Schrader (site-form) reconstruction of the transfer operator is unitarily conjugate — by the explicit $\sqrt{w}$ boundary dressing — to the symmetrised Dobrushin kernel; (ii) the *unnormalised Gibbs sums themselves* are exact matrix elements of the reconstructed operator's powers, $\mathrm{gibbsPathSum}\, w\, \beta\, N\, A\, B = \lambda^N \langle \hat{T}^N QA,\, QB\rangle$ and the partition function is the same shape at the dressed constant — identities, not bounds; and (iii) there is *one* mass $m > 0$ such that for *every* spatial extent L the projected operator norm is at most e^{-m} and every *mixed* connected correlator of the reconstructed operator obeys $\left|\langle u,\, \hat{T}^n v\rangle - \langle \Omega,\, u\rangle\langle \Omega,\, v\rangle\right| \leq \|u\|\,\|v\|\,(e^{-m})^n$, the zero-time case included — constants per-observable, rate alone uniform in the volume. All quantifiers live in the Lean statements; the prose here copies the kernel's cut. To situate the claim in the 2026 landscape of machine-checked constructive field theory: the abstract construction (reflection-positive data → physical Hilbert space → self-adjoint contractive transfer operator) has been machine-checked, sorry-free, in the Harvard formalization programme, *with the spectral gap carried as an undischarged hypothesis* (`GappedTransfer`); the Osterwalder–Schrader axioms have been machine-checked for the free field; and measure-side Dobrushin clustering has been machine-checked for lattice Yang–Mills. What we add, and what to our knowledge no machine-checked artefact yet contains, is the combination {concrete interacting model, gap *proved* rather than hypothesised, uniform in the spatial volume} on the reconstructed operator, *identified exactly against the raw Gibbs sums* — i.e. a concrete discharge of exactly the hypothesis shape the abstract frameworks carry. We do not claim Wightman reconstruction, a continuum limit, $SU(N)$, or any progress on the Clay problem.

1 Introduction and landscape

The Osterwalder–Schrader route builds a physical theory out of Euclidean data: a reflection-positive form gives a Hilbert space, and composition with the time shift gives a self-adjoint transfer operator. For a lattice slice system the construction is elementary — the content is

*With machine-checked artefacts produced on the programme's Colab plane; toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned to `0764272048...`. Repository: THE-ERIKSSON-PROGRAMME, branch `d3-closure`, module anchor commit `b47dda50e3a0b383b4830cab53d945694629ea46`; verification anchor **[SLOT: fresh-clone SHA, pending the final pass]**.

never the construction but the *spectrum*: a spectral gap at every fixed spatial extent L is classical (Perron–Frobenius; in this programme it is the earlier `coupled_gap_all_eigenvalues`), and it degenerates as $L \to \infty$ unless something uniform survives. What a thermodynamic limit can actually consume is *one* $m > 0$ valid for every L at once.

That uniform input exists in this repository: the Dobrushin corollary `dobrushin_ising_uniform_gap` produces, inside the window $2\tanh|\beta| + 2\tanh|\gamma| \leq \alpha < 1$, a single $m > 0$ bounding the projected transfer operator of the tilted kernel at every extent. This paper's contribution is the *identification*, in two stages. Stage one (Section 3): the operator that bound governs *is* — unitarily, by the explicit $\sqrt{w}$ boundary dressing, with no spectral-transport theory — the transfer operator forced on the reconstructed (site-form) theory. Stage two (Section 5): the *unnormalised Gibbs sums themselves* are exact matrix elements of that operator's powers — λ^N times an inner product, an identity valid at every real β and every positive weight, with the window entering only when the gap is invoked. The theorems are deliberately cheap: the same witnesses as D-6, no new estimate. Cheapness is the honesty of the claim — the uniformity is real because it is D-6's, and the reconstruction adds commuting squares, machine-checked.

1.1 Machine-checked prior work, cited before a referee does

The following table is load-bearing and its rows were collected by a four-modality search on 2026-08-04 (evidence: `docs/OS-R-PRIORITY-COLLATION.md` in the repository; the four raw agent reports are appended there verbatim).

Artefact	What is machine-checked	What is not
`mrdouglasny/reflection-positivity` (sorry-free ∼2026-06)	RP data → Hilbert space (completion/quotient) → self-adjoint contraction T; the dictionary $\langle [f], T^n[g]\rangle = \int f\,(g \circ \tau^n \circ \theta)\, d\mu$; trace formulas; susceptibility bounds *from* a gap	the gap itself: carried as the `GappedTransfer` *hypothesis* (`GroundGap.lean`); no spatial-volume-uniform statement; concrete instance is finite-lattice RP only
`OSforGFF` (arXiv:2603.15770)	all five GJ/OS axioms for the d=4 Gaussian free field, 0 sorries	reconstruction (listed as next milestone); any interacting model
`xiyin137/OSreconstruction`	Wightman-side GNS lane of the full OS↔Wightman theorem	the theorem: 53 sorries + 12 axioms at last README snapshot
`mrdouglasny/lgt` (complete 2026-05-04)	Chatterjee-style Dobrushin uniqueness for lattice YM, $U(n)$, with exponential clustering whose constants depend only on (n, d, β)	an operator gap; any reconstruction; the statement is per-volume on the measure side
`pphi2`	transfer-matrix spectral machinery for φ^4_2	rests on 27–29 project axioms
This paper	the $\sqrt{w}$-dressing unitary; the transported volume-uniform gap; the *exact* identities gibbsPathSum $= \lambda^N \langle \hat{T}^N QA,\, QB\rangle$ and $Z_N = \lambda^N \langle \hat{T}^N Q\mathbf{1},\, Q\mathbf{1}\rangle$; volume-uniform *mixed* clustering with the zero-time case (`os_reconstruction_measure_uniform`; core 8481, oracle 3026, 0 sorry — measured, Addendum 622)	Wightman axioms; continuum limit; $SU(N)$; Clay; an N-uniform denominator floor for the normalised quotient (the division-free six-term bound IS a theorem, Section 8)

There is a coordinated, active programme (Douglas et al., Harvard CMSA) whose declared live fronts are the OS reconstruction theorem, $P(\varphi)_2$, and YM mass gap at strong coupling, and whose framework carries the spectral gap as an undischarged hypothesis. The endpoint below is a concrete discharge of that hypothesis *shape* (different framework, same mathematical hole), which is why the comparison table above is printed in full before any claim wording.

2 The two geometries and the forced operator

Everything in this section is prior work of the repository, cited by module (the public trace of this lane is the ai.viXra `2607.009x` cluster); no new mathematics appears here. The slice system carries two pairings on complex boundary vectors: the *site* form, which weighs by $1/w$ (`siteForm`, module `SpatialReconstruction`), and the *bond* form carrying the decoupled kernel spatialKernel β (`bondForm`). The transfer operator is *forced*: $T = D_w K$ is the unique operator with $\mathrm{site}(u, Tv) = \mathrm{bond}(u, v)$ (`transferOp_unique`), self-adjoint for the site form at every β

(`transferOp_selfAdjoint`) and positive for $\beta \geq 0$; reflection positivity of the underlying measure is the lane's earlier bond/site pair (modules `SpatialReflection`, `SpatialOS`). On the measure side, the same repository carries the Gibbs bridge (module `SpatialGibbs`): the unnormalised two-point path sum `gibbsPathSum` and partition function `gibbsPartition` of the coupled slice system, with the in-tree operator bridge `gibbsPathSum_eq_iterate` (the path sum as an N-fold kernel iterate against dressed observables, dress $w\,A = \sqrt{w}\,A$). On the Dobrushin side it carries `symWeighted` $= \sqrt{w}K\sqrt{w}$ (module `PerronGap`), its normalisation `tiltKernel` $= \mathrm{symWeighted}/\lambda$ (module `DobrushinTilt`), the Euclidean packaging `opOf`, `vacOf`, `vacuumTransfer_opOf` (module `DobrushinTransport`), and the corollary `dobrushin_ising_uniform_gap` (module `DobrushinCorollary`). The seam fact this paper depends on — one `spatialKernel` declaration serving both geometries — is enforced by the elaborator in the module of Section 3.

2.1 The OS provenance: reflection positivity, collapse, null space, physical quotient

The operator of Section 2 is not an ansatz; its Osterwalder–Schrader provenance is machine-checked in the same tree, and we present it here because the reconstruction claim of this paper rests on it. All items below are green, in-tree prior work of the lane (modules `SpatialReflection`, `SpatialOS`, `SpatialReconstruction`); nothing in this subsection is new.

- *Reflection positivity over the measure.* The bond OS pairing of two half-chain observables is literally a sum over whole Gibbs paths — `osPairingBond_eq_gibbsSum`, through the splitting equivalence `bondEquiv` and the weight identity `gibbsWeight_joinBond` — and it is nonnegative on the diagonal for $\beta \geq 0$ (`osPairingBond_nonneg`; the restriction is sharp: `gibbsPathSum_neg_of_neg_beta` exhibits negativity at $\beta < 0$).

- *The collapse and its null space.* Half-chain observables collapse linearly onto boundary vectors (`collapseL`, surjective by `collapseL_surjective`), and the null space of the reflected site form is *exactly* the kernel of the collapse: $F \in \ker(\mathrm{collapseL}) \iff \mathrm{osPairingSite}\, F = 0$ (`mem_ker_collapseL_iff`). The quotient is by something identified, not by an abstract null space that happens to be there.

- *The physical quotient.* The quotient of half-chain observables by that kernel *is* the boundary space, as a bundled linear equivalence (`physicalEquiv`, via `quotKerEquivOfSurjective`).

- *The operator's provenance.* The transfer operator's matrix element between two collapsed half-chain observables is the reflected two-point sum of the Gibbs measure over whole paths, one slice further apart — `osPairing_transfer_gibbsSum`, stated for two observables, not one. Together with forcing (`transferOp_unique`), self-adjointness at every β (`transferOp_selfAdjoint`) and positivity for $\beta \geq 0$ (`transferOp_nonneg`), this is the OS construction of the operator whose spectrum Sections 4–7 control.

At finite extent the site form with $w > 0$ is definite, so on boundary vectors no further quotient intervenes; the quotient content lives one level up, on half-chain observables, where it is the `physicalEquiv` above. This is the honest division of labour between construction and spectrum in this model.

3 The dressing unitary

All statements of this and the following sections live in one module (`YangMills/OS/OSReconstructionUniform.le` sha256 `d9919e256ae59ac9...27bac149`, 20,428 bytes, ten-pass ladder on the plane, final `ELAB_EXIT 0`). Write $Qu := \sqrt{w}\,u$ for the boundary dressing of a real Euclidean vector (`qEmbed`; it coincides with the measure side's `dress`, complexified — `qEmbed_eq_dress` is definitional); the site form weighs by $1/w$.

Theorem 3.1 (A1, `siteForm_qEmbed`). *For $w > 0$ and real u, v:* $\mathrm{siteForm}\, w\,(Qu)\,(Qv) = \sum_\sigma u_\sigma v_\sigma$*. Per site the cancellation is elementary:* $\sqrt{w}u \cdot \sqrt{w}v \cdot (1/w) = uv$.

Theorem 3.2 (A2, `transferOp_qEmbed`). $T(Qu) = Q\big(\mathrm{act}\,(\mathrm{symWeighted}\, w\,\beta)\,u\big)$ *pointwise: the forced transfer operator on a dressed vector is the dressing of the symmetrised kernel's action.*

Theorem 3.3 (A3, `transferOp_iterate_qEmbed`). $T^{[n]}(Qu) = Q\big((\mathrm{act}\, S)^{[n]}u\big)$ *— the shape the in-tree measure bridge `gibbsPathSum_eq_iterate` consumes.*

Together with A1, the dressing is a unitary between the Euclidean ℓ^2 pairing and the site form at every finite extent (the site form with $w > 0$ is definite, so no quotient intervenes — this is the elementary, finite-volume case of the OS construction, and we say so plainly).

4 The transported volume-uniform gap

Theorem 4.1 (`os_reconstruction_uniform_gap`). *Let $\beta, \gamma \in \mathbb{R}$ and $0 < \alpha < 1$ with $2\tanh|\beta| + 2\tanh|\gamma| \le \alpha$. There is $m > 0$ such that for every L there are $\lambda > 0$ and $\Omega > 0$ with the fixed-point equation* $\sum_\tau \mathrm{tiltKernel}\,(\mathrm{sliceW}\,\gamma\, L)\,\beta\,\lambda\,\sigma\,\tau\,\Omega_\tau = \Omega_\sigma$*, the projected bound* $\|\mathrm{projectedTransfer}\,(\mathrm{opOf}\,(\mathrm{tiltKerne}$ e^{-m}*, and the A2 identity at* $w := \mathrm{sliceW}\,\gamma\, L$.

The proof is an `obtain` on `dobrushin_ising_uniform_gap` followed by A2: *the same witnesses*, no new analytic content — which is precisely the point. The uniformity is real because it is D-6's; the reconstruction contributes the identification, not a new estimate.

5 Exact identification against the Gibbs sums

The bridge from the raw measure to the operator is a chain of *identities* — no bound appears until the gap is invoked, and none of the statements below needs the window: they hold at every real β, every $\lambda \neq 0$, every positive weight.

Lemma 5.1 (B0b, `act_symWeighted_eq_smul_act_tilt`). $\mathrm{act}\, S\, u = \lambda \cdot \mathrm{act}\,(\mathrm{tilt})\,u$ *for $\lambda \neq 0$, and iterating (`act_symWeighted_iterate_eq_smul_tilt`, via the scalar transport lemmas `act_smul_fun`, `act_iterate_smul_fun`):* $(\mathrm{act}\, S)^{[N]}u = \lambda^N \cdot (\mathrm{act}\,\mathrm{tilt})^{[N]}u$.

Lemma 5.2 (B0a, `act_iterate_eq_opOf_pow`). $(\mathrm{act}\, M)^{[n]}u = (\mathrm{opOf}\, M)^n(\mathrm{toLp}\, u)$ *pointwise, for any kernel matrix M.*

Theorem 5.3 (B1, `gibbsPathSum_eq_inner_pow`). *For $w > 0$, $\beta \in \mathbb{R}$, $\lambda \neq 0$, $N \in \mathbb{N}$ and real observables A, B:*

$$\mathrm{gibbsPathSum}\, w\,\beta\, N\, A\, B \;=\; \lambda^N \big\langle (\mathrm{opOf}\,(\mathrm{tiltKernel}\, w\,\beta\,\lambda))^N\,(QA),\; QB\big\rangle,$$

an exact identity: the unnormalised Gibbs two-point sum is *λ^N times a matrix element of the reconstructed operator's powers, taken between dressed observables.*

Theorem 5.4 (B2, `gibbsPartition_eq_inner_pow`). *Under the same hypotheses, the partition function is the same shape at the dressed constant:* $\mathrm{gibbsPartition}\, w\, \beta\, N = \lambda^N \langle \hat{T}^N(Q\mathbf{1}),\, Q\mathbf{1}\rangle$.

Remark 5.5. *Dividing B1 by B2 cancels* λ^N*, so the* normalised *two-point function equals* $\langle \hat{T}^N QA,\, QB\rangle / \langle \hat{T}^N Q\mathbf{1},\, Q\mathbf{1}\rangle$ *— plain arithmetic from the two identities, recorded here as prose. The quantitative six-term* connected *bound is a theorem of the same module, in division-free form: Section 8. We print the boundary between identity and estimate exactly where the kernel puts it.*

6 Mixed clustering on a generic Hilbert space

The clustering machinery is stated for an arbitrary real Hilbert space H (a `section` with `[NormedAddCommGroup H] [InnerProductSpace ℝ H] [CompleteSpace H]`), so the three lemmas below are reusable against any transfer operator packaged as a `VacuumTransfer` — in particular they are exactly the consumer shape of the `GappedTransfer` hypothesis of the abstract frameworks in Section 1.1.

Lemma 6.1 (`norm_sub_inner_smul_le`). *For* $\|\Omega\| = 1$*:* $\|v - \langle \Omega,\, v\rangle\, \Omega\| \le \|v\|$ *(Pythagoras; removing the vacuum component does not grow the norm).*

Lemma 6.2 (`pow_apply_norm_le`). $\|S^n v\| \le r^n \|v\|$ *whenever* $\|S\| \le r$, $r \ge 0$.

Theorem 6.3 (`mixed_connCorr_bound`). *Let T be a `VacuumTransfer` with vacuum Ω, and suppose* $\|\mathrm{projectedTransfer}\, T\, \Omega\| \le r$ *with* $r \ge 0$*. Then for all* $u, v \in H$ *and all* $n \in \mathbb{N}$ including $n = 0$*:*

$$\left|\langle u,\, T^n v\rangle - \langle \Omega,\, u\rangle\, \langle \Omega,\, v\rangle\right| \;\le\; \|u\|\, \|v\|\, r^n.$$

The $n = 0$ case is the Pythagoras lemma; $n \ge 1$ goes through the in-tree `projected_pow_succ`.

This strictly extends the lane's earlier diagonal clustering (`os_reconstruction_uniform_clustering`, $u = v$, constant $\|v\|^2$), which remains in the module: two *different* observables, and the time-zero correlator is bounded too — the zero-time case is where a naive geometric-series argument is silent, and it is carried by the projection geometry instead.

7 The endpoint

Theorem 7.1 (`os_reconstruction_measure_uniform`). *Let* $\beta, \gamma \in \mathbb{R}$ *and* $0 < \alpha < 1$ *with* $2 \tanh|\beta| + 2\tanh|\gamma| \le \alpha$*. There is* one $m > 0$ *such that for* every *spatial extent L there are* $\lambda > 0$ *and* $\Omega > 0$ *with:*

(i) the fixed-point equation for $\mathrm{tiltKernel}\,(\mathrm{sliceW}\, \gamma\, L)\, \beta\, \lambda$ *at* Ω*;*

(ii) exact identification*: for every N and all real observables A, B,* $\mathrm{gibbsPathSum}\,(\mathrm{sliceW}\, \gamma\, L)\, \beta\, N\, A\, B = \lambda^N \langle \hat{T}^N(QA),\, QB\rangle$ *with* $\hat{T} = \mathrm{opOf}\,(\mathrm{tiltKernel} \ldots)$*;*

(iii) volume-uniform mixed decay*: for all u, v in the slice Hilbert space and all $n \ge 0$,* $\left|\langle u,\, \hat{T}^{\,n} v\rangle - \langle \mathrm{vacOf}\, \Omega,\, u\rangle\, \langle \mathrm{vacOf}\, \Omega,\, v\rangle\right| \le \|u\|\, \|v\|\, (e^{-m})^n$.

The witnesses of (i)–(iii) are `dobrushin_ising_uniform_gap`'s, consumed verbatim; the packaging is `vacuumTransfer_opOf` + `tiltKernel_symm`; (ii) is B1 at $w = \mathrm{sliceW}\, \gamma\, L$ (whose positivity is the in-tree `sliceW_pos`); (iii) is Theorem `mixed_connCorr_bound` at $r = e^{-m}$. One theorem, three clauses, no new constant: the mass that clause (iii) exhibits is the mass D-6 proved, now measured against the raw Gibbs sums through the exact identity of clause (ii).

8 The division-safe six-term connected bound

The one step the previous version named open is now a theorem, in two layers. The generic layer lives in the same arbitrary-Hilbert-space section as the mixed bound:

Theorem 8.1 (`six_term_connected_bound`). *Let T be a `VacuumTransfer` with vacuum Ω, let $0 \le r \le 1$ with $\|\mathrm{projectedTransfer}\, T\, \Omega\| \le r$, and write $F(u,v) := \langle u,\, T^N v\rangle$. Then for all q_A, q_B, q_1 and all N:*

$$\big|F(q_1,q_1)\, F(q_A,q_B) - F(q_A,q_1)\, F(q_1,q_B)\big| \;\le\; 6\, \|q_A\|\, \|q_B\|\, \|q_1\|^2\, r^N.$$

The proof is a ring identity in E-terms ($E(u,v) := F(u,v) - \langle \Omega,\, u\rangle\langle \Omega,\, v\rangle$): the pure product terms cancel *exactly*, six terms survive, each carrying at least one E factor bounded by Theorem `mixed_connCorr_bound`, and the two $E\cdot E$ terms spend one factor through $r^N \le 1$. The constant 6 is the honest count of surviving terms, not an optimised one.

The model layer consumes it at the dressed observables, *in raw Gibbs terms*:

Theorem 8.2 (`os_reconstruction_connected_uniform`). *Let $\beta, \gamma \in \mathbb{R}$ and $0 < \alpha < 1$ with $2\tanh|\beta| + 2\tanh|\gamma| \le \alpha$. There is* one *$m > 0$ such that for* every *spatial extent L there are $\lambda > 0$, $\Omega > 0$ with the fixed-point equation, and for every time depth N and all real observables A, B:*

$$\big|Z_N\, S_N(A,B) - S_N(A,\mathbf{1})\, S_N(\mathbf{1},B)\big| \;\le\; \lambda^{2N}\left(6\,\|QA\|\,\|QB\|\,\|Q\mathbf{1}\|^2\right)(e^{-m})^N,$$

where $S_N = \mathrm{gibbsPathSum}$, $Z_N = \mathrm{gibbsPartition}$, and $Q\cdot = \mathrm{toLp}(\mathrm{dress}\,\cdot)$.

This is the normalised connected-correlator bound multiplied through by the positive scale of Z_N^2: dividing by $Z_N^2 = \lambda^{2N}\langle \hat{T}^N Q\mathbf{1},\, Q\mathbf{1}\rangle^2$ recovers

$$\left|\frac{S_N(A,B)}{Z_N} - \frac{S_N(A,\mathbf{1})}{Z_N}\,\frac{S_N(\mathbf{1},B)}{Z_N}\right| \;\le\; \frac{6\,\|QA\|\,\|QB\|\,\|Q\mathbf{1}\|^2}{\langle \hat{T}^N Q\mathbf{1},\, Q\mathbf{1}\rangle^2}\,(e^{-m})^N,$$

which is printed here as prose arithmetic, not as a separate Lean statement. We do *not* claim an N-uniform denominator floor: at fixed L the even powers give $\langle \hat{T}^N Q\mathbf{1},\, Q\mathbf{1}\rangle \ge \langle \Omega,\, Q\mathbf{1}\rangle^2$, but the odd branch admits no such clean uniform bound without further work, and the division-free formulation above is exactly what the kernel proves. The boundary between identity and estimate — and between estimate and prose — sits where the kernel put it.

9 Reproduction

All numbers below are read from committed run artefacts; predictions were committed *before* their measurements (the programme's standing discipline — the ledger records nine consecutive exact count predictions for this lane before this paper's two).

Stage OS-R-1 (measured, ledger Addenda 616–619)

Quantity	Predicted	Measured
core jobs, unwired	8480 (ledger baseline)	8480
core jobs, wired	$8480 + 1$	8481
oracle reports, wired	$3010 + 7$	3017
`sorryAx` occurrences	0	0
endpoint axiom sets	standard triple	exactly `[propext, Classical.choice, Quot.sound]`

Fresh-clone witness at anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`: judges 110/110 in normal and `-O`; core 8481; oracle 3017; module sha256 bit-identical (Addendum 619; run cell `scripts/osr1_pass7_freshclone.py`).

Stage OS-R-2 (predictions Addendum 621, committed before the run; measurements Addendum 622)

Quantity	**Predicted (committed first)**	**Measured**
core jobs	8481 exact (file grew, no module added)	8481
oracle reports	$3017 + 9 = 3026$ exact	3026
`sorryAx` occurrences	0	0
new endpoint axiom sets	standard triple	exactly `[propext, Classical.choice,`
six-term stage (predictions Addendum 623, committed before the run):		
core jobs	8481 exact (file grew again)	**[SLOT: pass 15]**
oracle reports	$3026 + 2 = 3028$ exact	**[SLOT: pass 15]**
`sorryAx` occurrences	0	**[SLOT: pass 15]**
fresh-clone reproduction	bit-identical module, same counts	**[SLOT: final pass]**

The core count is the tenth consecutive exact count prediction of this lane: the module file grew from 12,347 to 20,428 bytes and the job total did not move. Module: `YangMills/OS/OSReconstructionUniform.l` final sha256 `3b3598ebcfde0ce5f2f6ba358e7e0607473ac66db7afbdd4a8b09e014db7f369` (32,533 bytes), banked at commit `b47dda50e` (measure stage: sha `d9919e25...`, 20,428 B at `3b0dfa9e8`) after the plane printed `ELAB_EXIT 0` on byte-identical source (the cell prints size and sha256 before elaborating; both matched the local file before banking). Pass history: passes 1–5 (OS-R-1, Addenda 614–616), passes 8–10 (OS-R-2, Addendum 621: the inner-product notation trap at this Mathlib pin, then a single `nlinarith` orientation failure in the Pythagoras lemma, then green). The wiring unit with the registered predictions is committed as `scripts/osr2_pass11_wiring.py` *before* its run, at commit `520e177a7`.

10 What is not claimed

No Wightman axioms, no OS→Wightman analytic continuation, no continuum limit, no $SU(N)$, no statement about the Clay problem. The window is the Dobrushin window; outside it the programme's own kill-test (repository, Addendum 610) shows the standard-cone Birkhoff route is projectively blind to the site weight, which is why no claim beyond the window appears here. The model is the $\mathbb{Z}_2$ Ising slice; "reconstruction" means the elementary finite-volume site-form construction (the quotient content lives on half-chain observables, Section 2.1), and the theorems are about its *spectrum and its exact relation to the Gibbs sums*, not about the construction being difficult. The connected bound is printed in its division-free form (Section 8); what is NOT claimed is an N-uniform lower bound for the denominator $\langle \hat{T}^N Q\mathbf{1}, Q\mathbf{1} \rangle$ at fixed L — the even branch has one, the odd branch is not proved here, and the prose does not paper over the difference.

References

[1] M. R. Douglas, S. Hoback, C. Mei, Y. Nissim, *Formalization of QFT*, arXiv:2603.15770 (2026). Repositories: `mrdouglasny/OSforGFF`, `mrdouglasny/reflection-positivity`, `mrdouglasny/lgt`, `mrdouglasny/pphi2`; `xiyin137/OSreconstruction`.

[2] K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973), **42** (1975).

[3] J. Glimm, A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer (1987).

[4] R. L. Dobrushin, *The description of a random field by means of conditional probabilities and conditions of its regularity*, Theory Probab. Appl. **13** (1968).

[5] L. Eriksson, *From the Gibbs Weight to the Spectral Gap: A Complete Machine-Checked Osterwalder–Seiler Chain for the $\mathbb{Z}_2$ Lattice Gauge Chain*, ai.viXra:2607.0078 (2026) — the zero-extension, explicitly non-volume-uniform predecessor of this paper.

[6] L. Eriksson, *the dobrushin-matrix paper of the same repository* (`papers/dobrushin-matrix/`), source of `dobrushin_ising_uniform_gap` (D-6, ledger Addendum 603).